

Evaluating the task value of budgeted reobservation for UAV disaster inspection

Author names to be confirmed

Affiliations and corresponding-author details to be confirmed,

Abstract

Aerial disaster inspection requires deciding whether another observation is worth its cost. A closer view may improve damage classification without improving an inspection answer. We study this distinction in a single-UAV imagery simulation with a fixed camera model and budgeted reobservation. Paired predictions for 3,753 buildings in 44 regions quantify correction opportunities and harmful changes. At a reobservation fraction of 0.25, calibrated-entropy allocation raises macro-F1 from 0.6118 to 0.6433, compared with 0.6241 for random allocation and 0.6448 for raw entropy. Full reobservation corrects 191 decisions but harms 126. In 160 online inspection questions, configurations answer 48–51 correctly, versus 50 without reobservation. The expected-entropy policy spends all 15 additional observations on eight already-correct, no-damage questions from one event. These records reveal a mismatch between changing local evidence and directing observations to unresolved tasks. The findings are conditional on a perception backbone with prior evaluation-event exposure. Unequal online action counts and incomplete execution provenance further limit causal interpretation. The study separates observation change, uncertainty change, and task value through reproducible analysis of recorded evidence, without establishing unseen-event generalization or a language-model reliability guarantee.

Keywords: UAV disaster inspection, Embodied perception, Active perception, Reobservation budget, Uncertainty calibration, Vision-language agents

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1. Introduction

An aerial inspection agent may locate a damaged area and still lack the evidence needed to answer a specific question about it. A wide view supports regional coverage, whereas a closer view allocates more image pixels to a smaller area. Moving the sensor therefore changes both the available detail and the context retained for interpretation. In disaster inspection, this trade-off matters because questions about individual buildings, damage counts, and the spatial distribution of severe damage require different evidence. An additional observation is useful only when its contribution to the requested decision justifies the observation cost.

Aerial vision-and-language navigation has developed datasets and agents for following instructions through outdoor environments. AerialVLN and CityNav establish substantial navigation settings, while OpenFly provides a platform with trajectories spanning multiple heights and scenes [1, 2, 3]. These developments make motion-conditioned visual reasoning a concrete research problem. Navigation success, however, does not directly measure whether a new view improves a disaster assessment. Building-damage data such as xBD [4] support evaluation of visual classification, but a change in building-level accuracy alone does not reveal which inspection questions benefit from an agent’s sensing decisions.

Foundation models provide a practical interface between task language, visual evidence, and tools. Modular navigation systems, hierarchical semantic planning, and aerial agent frameworks already combine these components [5, 6, 7]. CityEQA connects active urban exploration to question answering [8], while disaster interpretation agents perform sequential analysis of remote-sensing images [9]. Their capabilities motivate a narrower question here: when observations are limited, do uncertainty-based reobservation rules direct the sensor toward evidence that changes task outcomes? This question cannot be answered by the existence of an action–observation loop or by an increase in a model’s self-reported confidence. It requires paired outcomes and an account of where the observation budget was spent.

We address this question using DisasterClaw, a single-UAV inspection implementation operating over georeferenced disaster imagery. The study separates a controlled building-level experiment from an online question-answering experiment. In the first, each building has paired predictions at three camera footprints, allowing observation selection to be compared under a common building budget. In the second, the agent receives a rendered image and structured evidence, can request additional observations, and returns an answer or abstention. This separation is necessary: the offline allocator ranks a collection of buildings, whereas the online controller decides whether to

inspect again within one question. They use related uncertainty signals but solve different allocation problems.

The contributions are threefold. First, we specify an imagery-based observation model and an evaluation design that separates changes in visual classification from changes in inspection answers. Second, we reconstruct budget-dependent performance from paired predictions and distinguish beneficial from harmful reobservations, with exploratory uncertainty estimates that retain within-region dependence. Third, we trace online observation use to question types and outcomes, identifying a concrete concentration of actions on already-correct cases. These are mechanism and evaluation contributions within the recorded setting. The available experiments do not support a new optimal controller, a state-of-the-art navigation claim, or generalization to disasters unseen by the perception backbone.

2. Related work

2.1. *Language-conditioned aerial autonomy*

Vision-and-language navigation connects instructions to sequential visual decisions [10]. In aerial environments, AerialVLN extends this task to UAV navigation, and CityNav emphasizes real-world aerial navigation data [1, 2]. OpenFly includes a multi-engine simulation platform, a collection toolchain, and height-varying navigation trajectories [3]. Thus, altitude variation and language-conditioned aerial movement are established research directions. Our experiment focuses on the value of revisiting visual evidence for damage assessment, rather than proposing another navigation benchmark or comparing success rates across incompatible environments.

Large pretrained models can be composed into navigation systems without treating every component as a newly learned policy [5]. CityNavAgent uses hierarchical semantic planning and global memory, while spatial representations have been studied specifically for language-model reasoning in aerial navigation [6, 11]. AerialClaw further develops modular aerial agents with skills and feedback [7]. These works constrain the novelty claimed for DisasterClaw: semantic planning, spatial memory, and tool execution provide implementation support. The scientific question examined here concerns the allocation and measured utility of additional observations.

2.2. *Active perception and decision value*

CityEQA directly connects urban exploration to question answering through a hierarchical agent [8]. Its Collector adjusts the observation pose, and its evaluation examines answers across refinement steps. Active aerial question answering and the effects of view refinement therefore already have

direct precedents. Our narrower focus is paired correction and harm under a common building-level observation budget, followed by a separate diagnosis of observation use in disaster questions. The different environments and scoring protocols preclude a direct performance ranking against CityEQA.

Active perception treats sensing as an activity that can be controlled to serve a task [12]. More recent planning under partial observability combines information gathering with uncertainty about states and action outcomes [13]. This literature motivates evaluating an observation by the decision it enables, rather than by visual change alone. The present study uses a simpler setting: two-dimensional source imagery, a finite camera ladder, and short inspection episodes. It evaluates heuristic observation rules and their failure cases; it does not implement a general belief-space planner.

Entropy reduction is a convenient signal for these rules, but it is not the same quantity as expected task utility. A probability distribution can become more concentrated around an incorrect class. Conversely, an observation may improve a count or spatial answer without changing the most uncertain building’s label. We therefore retain separate endpoints for classification, correction and harm, final question correctness, and executed observation count. The distinction is operational, and does not require a theorem equating prediction flips with all possible forms of observation value.

2.3. Disaster interpretation and uncertainty

xBD provides paired satellite imagery and building-damage annotations for disaster assessment [4]. RescueADI studies autonomous agents for adaptive interpretation of remote-sensing disaster images [9]. These resources and systems establish that both damage recognition and tool-based disaster analysis predate this work. Our setting adds a controlled change in rendered camera footprint and asks whether the associated evidence changes justify a limited sensing budget. Because the source images are satellite products, this is an emulation of aerial inspection rather than a collection of physical UAV revisits.

Temperature scaling addresses probability calibration, while evaluation under dataset shift shows why in-distribution calibration should not be treated as a universal reliability property [14, 15]. Selective prediction measures the trade-off between coverage and error, and uncertainty-aware language-model planning has also been studied through help-seeking behavior [16, 17]. We use these distinctions to separate calibrated detector probabilities, prediction-set triggers, and the VLM’s answer confidence. A calibration temperature does not establish a guarantee for the composed agent, and a prediction-set size used as a sensing score does not establish coverage under the present event shift and adaptive observation process.

3. Problem formulation

3.1. Inspection task and observation model

A task consists of a question q , a region of interest (ROI), and a reference answer y_q . A single UAV has an observation state $s_t = (x_t, y_t, h_t)$ describing its map position and height. Its image is $I_t = \mathcal{R}(\mathcal{M}, s_t)$, where $\mathcal{M}$ is the available georeferenced imagery and $\mathcal{R}$ crops and resamples a ground footprint. The notation describes the implemented observation state, not a complete aircraft dynamics model. Question types cover building damage, damage counts, presence of severe damage, and spatial relations.

For a nadir-view camera with horizontal field of view θ and output width W , the simulated footprint width L and effective ground sampling distance g are

$$L(h) = 2h \tan(\theta/2), \quad g(h) = \frac{L(h)}{W}. \quad (1)$$

We use $\theta = 60^\circ$, $W = 1024$, and an assumed source scale of 0.5 m/pixel. A source tile therefore spans 512 m. Footprints of three, two, and one tile widths give cruise, intermediate, and floor views at approximately 1330.2, 886.8, and 443.4 m, with effective GSDs of 1.5, 1.0, and 0.5 m/pixel. The heights follow from the chosen camera parameters and source-scale assumption. They do not validate that scale for every source image or establish aircraft operating limits.

Shrinking the footprint increases the pixels assigned to the target ROI while removing surrounding context. At the floor, the image reaches the assumed source resolution; further physical detail cannot be recovered from this imagery. The renderer does not model three-dimensional occlusion, new viewing angles, atmospheric effects, motion blur, or changes between real acquisition times. Figure 1 summarizes the two evaluation paths supported by this observation model.

3.2. Budget and task endpoints

In the offline experiment, N buildings each have a cruise probability vector p_i^c and a floor vector p_i^f over four damage classes. For budget fraction b , a policy selects a set S_b of $k = \text{round}(bN)$ buildings. The evaluated vector is

$$p_i^{(b)} = \begin{cases} p_i^f, & i \in S_b, \\ p_i^c, & i \notin S_b. \end{cases} \quad (2)$$

Each selected building counts as one additional observation in this abstraction. This accounting does not represent a route that physically visits every selected building, nor does it price the distance between them.

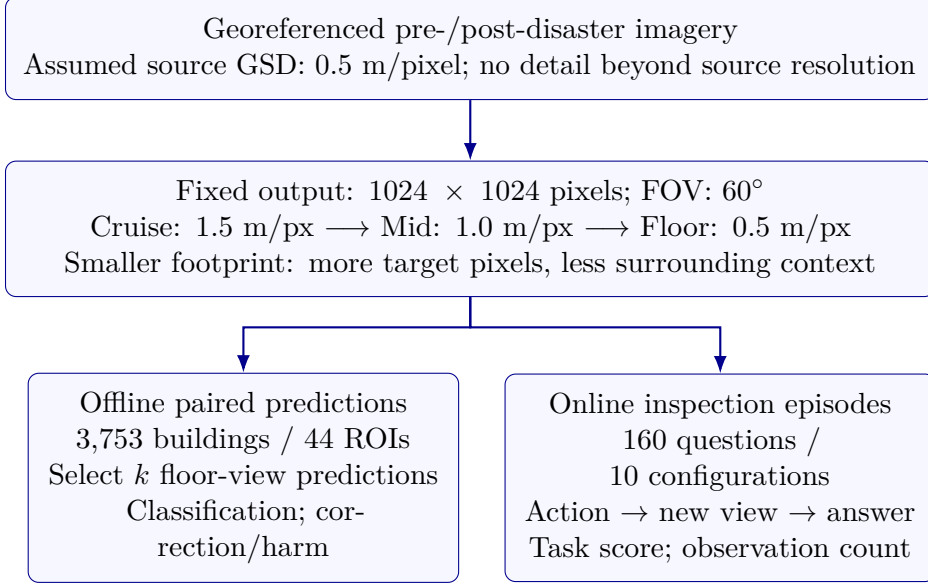


Figure 1: Observation model and two evaluation paths. The camera ladder changes the crop and resampling of existing imagery. Offline allocation selects paired building predictions under a common budget; the online controller acquires another rendered view within a question episode. The paths have different units of allocation and distinct final endpoints.

Let $\hat{y}_i^v = \arg \max_j p_{ij}^v$. A correctable case has $\hat{y}_i^c \neq y_i$ and $\hat{y}_i^f = y_i$; a harmful case has $\hat{y}_i^c = y_i$ and $\hat{y}_i^f \neq y_i$. For a fixed selected set, the accuracy difference is exactly

$$\Delta \text{Acc}(S_b) = \frac{N_{\text{corrected}}(S_b) - N_{\text{harm}}(S_b)}{N}. \quad (3)$$

This accounting identity applies to the paired hard-label endpoint. It is not a characterization of macro-F1, probability quality, or language-task utility. An unchanged predicted label may still accompany changes in those quantities.

Online, the agent can search, reobserve by centering or descending, answer, or abstain, subject to configured episode limits. We evaluate question correctness over all questions, counting abstentions as unsuccessful answers, and report actual reobservation actions separately. If a_q is the final answer, the task score is $|Q|^{-1} \sum_q \mathbf{1}[a_q = y_q]$. The purpose is to assess whether extra sensing changes this endpoint and where its cost is incurred. No optimization guarantee is assumed for the implemented controller.

4. Inspection implementation and observation policies

4.1. Perception and task grounding

The implementation combines a damage-perception backend, a georeferenced semantic representation, a language-conditioned inspection controller, and movement tools. The evaluated backend is recorded as `xview2_first`, with a single ResNet-34 configuration and seed 0. It processes pre-/post-disaster image pairs and supplies detected building locations and four-class damage probabilities. The four classes are no damage, minor damage, major damage, and destroyed. This backend is shared within the principal offline comparisons; its training-exposure limitation is stated in Section 5.

In the offline renderer, the post-disaster footprint is first sampled at the fixed camera output resolution. The corresponding pre-disaster window is rendered at the assumed source scale. The post-disaster image is then resized to the pre-disaster input dimensions for the paired-image detector; this resizing adds no new information. Thus, the ladder primarily degrades the post-disaster observation while preserving a higher-resolution archival reference. It should not be interpreted as simultaneous camera degradation of both images.

Predictions inside the target ROI are matched to annotated buildings by a greedy, one-to-one nearest-centroid rule, using a distance threshold of 12 m. An unmatched building receives a no-damage probability vector before normalization and temperature scaling. This convention is material: the reported building endpoint combines localization, matching, and damage classification, and may reward missed detections of undamaged buildings. We retain it when reproducing the archived experiment, disclose unmatched counts, and avoid presenting the scores as pure classification on perfectly localized buildings.

For online inspection, the controller receives the current rendered image together with structured counts and a semantic-map snapshot. A damage question can include a marked target location and a finite answer vocabulary. Spatial representations and memory support grounding and search; they are engineering components informed by prior aerial planning and representation work [6, 11]. The evaluated episodes do not isolate their contribution. The system’s ability to expose these tools must therefore be distinguished from an experimentally established navigation improvement.

4.2. Offline allocation rules

Let r_{ij} denote an uncalibrated class probability. Temperature scaling produces

$$p_{ij}(T) = \frac{r_{ij}^{1/T}}{\sum_{\ell=1}^4 r_{i\ell}^{1/T}}, \quad H(p_i) = -\frac{1}{\log 4} \sum_{j=1}^4 p_{ij} \log p_{ij}. \quad (4)$$

The implementation normalizes probabilities, clips small values for numerical stability, and rounds normalized entropy to three decimal places for ranking. A positive scalar temperature preserves each vector’s highest-probability class, but can change the ordering of uncertainty between buildings. All evaluated allocation outputs use the same calibrated vectors in Eq. (2). The raw-entropy rule changes the ranking score, not the probability representation used to score the final predictions.

The no-reobservation rule always retains the cruise prediction. Random allocation samples k buildings without replacement. Raw and calibrated entropy select the k largest corresponding cruise entropies. The expected-entropy-reduction rule fits the mean paired difference $H(p^c) - H(p^f)$ on development buildings, conditioned on the cruise predicted class and one of five equal-width cruise-entropy bins. Empty class/bin cells fall back to the class mean and then the global mean. This is an empirical uncertainty-reduction proxy; it is not trained on task correctness or the cost of an online trajectory.

The prediction-set rule sorts classes by probability and counts how many are needed for their cumulative probability to reach the frozen adaptive prediction set (APS) threshold. This rule draws on adaptive conformal classification [18], but uses the resulting set size only as a sensing priority here. Finally, a label-informed diagnostic assigns score +1 to correctable cases, −1 to harmful cases, and zero otherwise. It uses evaluation labels and cannot be deployed. Because it ranks accuracy changes rather than optimizing macro-F1, its curve is not a theoretical macro-F1 upper bound. Ties in deterministic scores retain input order. The random curve uses seed 0 and independent draws at each nonzero budget, so its selected sets are not nested across budget fractions.

4.3. Online triggering and motion feedback

The online controller evaluates whether to reobserve after producing a candidate answer and associated perception evidence. A trigger can use raw or calibrated entropy, a prediction set containing more than one class, a fixed allocation, or an expected-entropy lookup. The configured entropy threshold

is 0.7. For the expected-entropy trigger, the controller computes

$$\tilde{G}_t = \max\{0, H_t - \hat{\mu}(g_{t+1}, \hat{c}_t)\}, \quad (5)$$

and requests reobservation when $\tilde{G}_t > 0.02$, subject to available budget and action checks. Here $\hat{\mu}$ is a development-set mean entropy indexed by GSD and predicted damage class. The current implementation requires the compatible FOV entropy-table schema for this mode.

This lookup differs from the paired offline predictor. Its stored mean at a future GSD groups development examples by the class predicted at that GSD, whereas the online query supplies the current predicted class. Consequently, Eq. (5) is a plug-in heuristic with a conditional-variable mismatch when classes change across views. It is not an estimate of expected question utility given the current history. This distinction is especially relevant when a local damage signal controls reobservation for a count or spatial question.

An accepted reobservation invokes the movement interface to center on evidence, descend within the camera ladder, or combine both operations. The next rendered view is passed through perception and answer generation. The controller limits the episode and records the candidate answer, uncertainty, decision, action, and remaining budget. Answer validity and missing-target conditions can lead to abstention or further search. In the inspected implementation, a valid answer is not rejected solely because its confidence lies below a configured answer-confidence threshold. We therefore do not describe the controller as implementing calibrated selective prediction; any confidence-threshold analysis below is a retrospective evaluation of recorded answers.

5. Experimental setup

5.1. Data populations and exposure boundary

The development population contains 2,921 buildings from 20 ROIs in Hurricane Harvey and the Mexico earthquake. The principal evaluation population contains 3,753 unique buildings from 44 ROIs: 1,303 buildings in Hurricane Michael, 1,448 in the Moore tornado, and 1,002 in Nepal flooding. There are 15, 15, and 14 evaluation ROIs in these events, respectively. The development and evaluation event sets are disjoint for fitting the allocation statistics. This split does not imply an event-disjoint perception model: the frozen manifest explicitly marks the `xview2_first` backbone as having prior evaluation-event exposure. All principal offline and online results are interpreted within that boundary.

The online population contains 160 unique questions across the same 44 evaluation ROIs, with 58, 47, and 55 questions from the three events,

respectively. The question-type counts are 44 presence, 43 damage, 43 count, and 30 spatial questions. Ten configurations were evaluated on an identical set of question IDs, producing 1,600 records without duplicate configuration/question pairs. These are repeated evaluations of 160 questions, not 1,600 independent tasks. Building predictions and question records have different units and cannot be pooled into one sample size.

The archived results permit arithmetic reconstruction from predictions, labels, and correctness flags. The source image collection, frozen question-set entities, and exact training data are not all available in the current reproduction package. Thus, reconstruction verifies the reported arithmetic and local file consistency, rather than independently certifying annotations, checkpoint exposure, or the original scoring implementation. The outstanding requirements are stated in Appendix A.

5.2. Configurations and recorded execution

The frozen calibration parameters are $T = 3.5635948726$ and APS threshold $\hat{q} = 0.8064657026$, with nominal calibration parameter $\alpha = 0.1$. The temperature was selected on the development cruise probabilities by minimizing negative log-likelihood over 241 logarithmically spaced candidates in $[0.25, 4]$. The offline allocation budgets are $b \in \{0, 0.1, 0.25, 0.5, 1\}$; $b = 0.25$ is the frozen reference operating point. We reproduce all seven policy curves, retaining the label-informed diagnostic only for mechanism interpretation.

The online configurations comprise hold, random allocation, fixed allocation, raw entropy, calibrated entropy, APS triggering, expected entropy reduction, and three motion variants. The nominal limits allow six search actions and up to two reobservations per question, with hold allowing no reobservation. Random and fixed controls were intended to derive budgets from the expected-entropy reference. The three motion variants allow centering, descent, or combined motion. Their actual costs are reported alongside correctness because the archived budget check fails: intended matching and executed matching differ.

The online runs use seed 42 and two shards. Their manifests record Python 3.11.15, NumPy 2.4.4, PyTorch 2.11, and Transformers 5.5.4, with zero reported execution errors. Both point to the same frozen manifest and question-set hash. However, they identify a dirty source tree, and the model-identity fields are empty. The software versions are recorded execution metadata, not a validated environment reconstruction.

Author query: Supply the exact VLM provider, model/version, decoding parameters, archived prompt, and the uncommitted source snapshot used for these runs. Model identity cannot be inferred from an empty manifest field.

5.3. Metrics and statistical scope

For the building endpoint we report accuracy, unweighted macro-F1 over four classes, multiclass Brier score, negative log-likelihood (NLL), and expected calibration error (ECE) using 15 equal-width confidence bins. Brier score is the mean sum of squared differences from the one-hot label, without division by the number of classes. Corrected and harmed counts follow Eq. (3). These metrics are computed on the same building population, including unmatched cases under the archived convention.

We add an exploratory paired ROI bootstrap for two comparisons at $b = 0.25$: calibrated entropy versus random allocation, and calibrated versus raw entropy. Each of 2,000 resamples draws ROIs with replacement within each observed event, preserving that event’s ROI count. We aggregate the sampled confusion matrices before recomputing macro-F1 differences. The seed is 20260905; the reported limits are the 2.5th and 97.5th percentiles. Allocation masks, fitted parameters, and the observed event mixture are held fixed. These intervals address sampling variation among ROIs within the three events, not training uncertainty, repeated random-policy seeds, or transfer to new disasters. The analysis was added after inspecting the records and is reported as exploratory.

Online accuracy counts every question in its denominator. We also report gain and loss relative to hold on paired question IDs, actual additional observation counts, and the number of episodes receiving them. Differences between configurations may include generation variability and unmatched exposure to sensing; they are not automatically caused by motion. For confidence analysis, we evaluate non-abstained answers at fixed retrospective thresholds and retain all answers tied at each threshold. Coverage is accepted questions divided by 160, and risk is the error fraction among accepted questions. No threshold is fitted or deployed by this analysis.

Motion counts are the principal online cost measure. The implementation also estimates movement time from geometric displacement and configured speeds. It lacks a validated aircraft energy model, realistic battery consumption, and aerodynamic constraints. We therefore do not interpret this time estimate as measured flight endurance or use it to claim energy efficiency, safety certification, or fixed-wing feasibility.

6. Results and discussion

6.1. The camera ladder changes the building endpoint

Table 1 reports the reconstructed final evaluation population, rather than development-set scores. Moving from cruise to the floor increases macro-F1 from 0.6118 to 0.6542 and accuracy from 0.7250 to 0.7423. Brier score and NLL

also decrease. The camera ladder therefore produces a measurable response in this perception pipeline. This response is conditional on the existing imagery, matching convention, and exposed backbone; it is not evidence from real changes in UAV acquisition altitude.

Table 1: Building-level results at three simulated views on the same 3,753 evaluation buildings. Lower Brier score and NLL are better.

View	GSD (m/px)	Acc.	Macro-F1	Brier	NLL
Cruise	1.50	0.7250	0.6118	0.4814	1.3564
Mid	1.00	0.7357	0.6327	0.4669	1.3314
Floor	0.50	0.7423	0.6542	0.4616	1.3073

Across cruise and floor, 383 buildings change predicted class. Of these, 191 change from incorrect to correct and 126 from correct to incorrect; the other 66 change between incorrect classes. The net accuracy gain is therefore 65/3,753, or 1.73 percentage points. Full reobservation does not simply reveal a uniformly better label: it exchanges some correct judgments for incorrect ones. A useful selection rule must identify beneficial opportunities while avoiding harmful changes.

Localization also affects this endpoint. The cruise, intermediate, and floor records contain 912, 916, and 907 unmatched buildings, respectively. Among the 191 corrected cases, 35 involve an unmatched record in at least one of the two views; among the 126 harmed cases, 18 do so. The remaining 156 corrections and 108 harms involve detections in both views. These counts show why the observed change should be attributed to the full perception-and-matching pipeline rather than solely to a damage classifier receiving sharper crops.

An added sensitivity analysis restricts evaluation to the 2,757 buildings with matched detections in all three views. Macro-F1 still rises, from 0.7069 at cruise to 0.7489 at the floor (Table A.1). Thus, the direction of the view response persists when cases involving an unmatched record are removed. This is a selected subset, conditioned on outcomes of the detection pipeline; its higher absolute score is not an improved estimate of full-population performance.

6.2. Budgeted allocation and uncertainty quality

At $b = 0.25$, each active offline allocation rule selects 938 buildings (Table 2). Calibrated entropy obtains macro-F1 0.6433, compared with 0.6241 for random allocation and 0.6118 without reobservation. It produces 53 net additional correct predictions, versus 19 for random allocation. Thus,

uncertainty-based selection can identify useful reobservations in this population under a common building budget.

Table 2: Offline allocation at $b = 0.25$. All active rules select 938 floor-view predictions and are scored with the same calibrated probability vectors. Net correct counts corrections minus harms relative to cruise. The label-informed row uses evaluation labels.

Policy	Actions	Acc.	Macro-F1	ECE	Net correct
No reobservation	0	0.7250	0.6118	0.2423	0
Random	938	0.7301	0.6241	0.2437	19
Raw entropy	938	0.7397	0.6448	0.2578	55
Calibrated entropy	938	0.7391	0.6433	0.2609	53
Expected entropy reduction	938	0.7357	0.6308	0.2551	40
APS set size	938	0.7343	0.6367	0.2549	35
Label-informed diagnostic	938	0.7759	0.6928	0.2853	191

The strongest practical macro-F1 at this budget is obtained by raw entropy (0.6448), slightly above calibrated entropy. Expected entropy reduction reaches 0.6308 and APS set size reaches 0.6367. The evidence therefore does not establish a benefit from calibrating the allocation score or from replacing uncertainty with the fitted entropy-reduction proxy. Entropy reduction is an imperfect target because becoming more confident need not correct the predicted class. Its fitted conditional means also average across buildings whose errors and visual contexts differ.

Figure 2 shows the full budget response. All active policies converge to the same floor endpoint at $b = 1$, because every building then uses its floor prediction. The label-informed diagnostic can exceed that endpoint at intermediate budgets by preferentially selecting corrections and postponing harmful cases. This behavior illustrates the importance of selection; it is not achievable without label information and is not an upper bound for arbitrary task reasoning.

The exploratory ROI bootstrap gives a calibrated-entropy minus random macro-F1 difference of 0.0192, with interval $[0.0097, 0.0419]$. The corresponding difference from raw entropy is -0.0015 , with interval $[-0.0093, 0.0061]$. These estimates support a conditional distinction between entropy selection and the recorded random allocation, while leaving no clear advantage for calibration over raw entropy. They do not establish equality of those two policies or quantify event-level generalization.

Probability quality does not follow the same ordering as macro-F1. At $b = 0.25$, calibrated-entropy allocation raises ECE from 0.2423 to 0.2609 and NLL from 1.3564 to 1.3769, although Brier score decreases from 0.4814 to 0.4763. Selectively replacing views can change the confidence distribution as

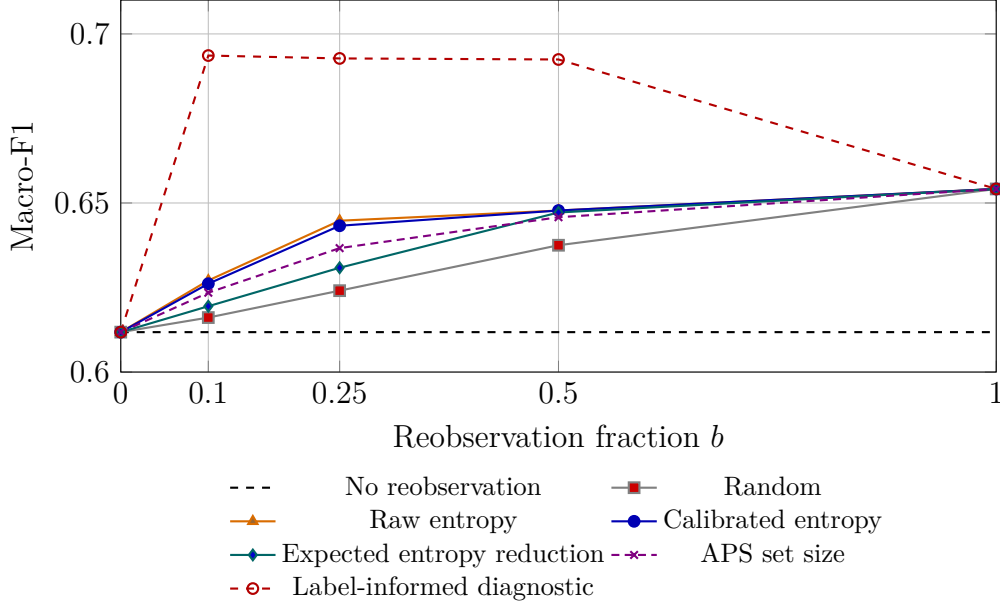


Figure 2: Offline macro-F1 against reobservation fraction on the fixed evaluation population. Lines connect the five evaluated budgets; random selections are not nested. The label-informed curve is a retrospective accuracy-oriented diagnostic.

well as the errors. Consequently, the improvement in hard-label performance cannot be described as an overall calibration improvement.

6.3. Offline gains do not establish an online task gain

Table 3 summarizes all recorded online configurations. Hold answers 50 of 160 questions correctly. Random and center-only configurations answer 51 correctly, expected entropy reduction answers 49, and the complete range is 48–51. These differences are only one or two questions relative to hold. The records provide no stable demonstration that additional sensing improves the final inspection endpoint.

The costs are unequal. Expected entropy reduction executes 15 reobservations in eight episodes, random executes 14, fixed executes 15, and APS executes 44 in 29 episodes. Each of the three motion variants executes 11. Their intended reference matching cannot be assumed from configuration names. A record-level check identifies three questions where each motion variant’s allocated budget differs from the current reference, accounting for four missing actions. The runs permit crash-resume, and the affected variant records predate the current reference records. This is consistent with stale dependent records surviving a resumed evaluation, although the complete

Table 3: Recorded online outcomes on 160 paired questions. Actions are executed re-observations; episodes count questions receiving at least one. Gain/loss compares final correctness with hold and does not identify the causal effect of an action. Motion variants and random allocation did not match the reference’s actual budget.

Configuration	Correct	Actions	Episodes	Gain	Loss
Hold	50/160	0	0	0	0
Random	51/160	14	14	1	0
Fixed	49/160	15	8	1	2
Raw entropy	49/160	15	8	1	2
Calibrated entropy	49/160	15	8	0	1
APS trigger	50/160	44	29	2	2
Expected entropy reduction	49/160	15	8	1	2
Center only	51/160	11	6	1	0
Descend only	48/160	11	6	1	3
Combined motion	49/160	11	6	0	1

execution history is unavailable. The variants remain descriptive observations; they do not provide a clean causal decomposition of centering and descent.

The limited task gain also cannot be inferred directly from the building-level result. The offline score aggregates many building decisions, while each online question may depend on a target marker, semantic counts, direction reasoning, or an answer format. A higher building macro-F1 need not repair the evidence required by a particular question. Moreover, comparison with hold includes differences in generated answers even when no additional observation occurs. The one-question loss in count performance for expected entropy reduction occurs with zero count-question reobservations (Table 4); it cannot be attributed to a descent harming the view.

Table 4: Question-type breakdown for hold and expected entropy reduction. Correct counts use all questions of each type. All expected-policy sensing actions are concentrated in damage questions.

Question type	Questions	Hold correct	Expected correct	Expected actions
Count	43	17	16	0
Damage	43	14	14	15
Presence	44	15	15	0
Spatial	30	4	4	0

6.4. A trace-level failure mode: sensing already-correct cases

The expected-entropy policy reobserves eight questions, all from the Moore tornado, all asking about damage, and all with the no-damage reference answer.

Every one is already answered correctly, and none changes its answer after reobservation. The calibrated-entropy policy exhibits the same concentration. Their extra observations therefore do not target the unsuccessful count and spatial judgments that contribute to the low overall accuracy. This is a concrete failure of task targeting in the recorded episodes, rather than a general claim that another view cannot help disaster inspection.

The APS trigger covers a broader set: its 29 reobserved episodes span three events and four question types, and 16 of them finish correctly. Its greater action count nevertheless leaves overall correctness at 50/160. More widespread triggering alone is therefore insufficient in this sample. A future controller should connect the requested evidence to the question’s unresolved requirements, but the present data do not evaluate such a replacement policy.

Across configurations, 93 reobserved episodes contain comparable initial and final confidence values. Confidence rises in 74, is unchanged in 18, and falls in one; the mean increases from 0.8344 to 0.9463. These are repeated configuration/question records, not 93 independent tasks. Because many selected episodes were already correct, this increase cannot by itself demonstrate additional task success. Nor does it show that confidence is always made less reliable: that requires probability-quality measurements on a defined population, rather than a conclusion drawn from confidence movement alone.

6.5. *Confidence and implications for aerospace autonomy*

Figure 3 gives retrospective risk–coverage operating points for hold, expected entropy reduction, and APS triggering. The curves retain all confidence ties and exclude already-abstained outputs from the accepted set. Their points represent different selected question populations; connecting them is a visual aid, not evidence that intermediate coverage levels are attainable without breaking ties. High-confidence answers can still be wrong, and small high-threshold subsets can appear reliable because of their composition. The curves do not validate an operational reject threshold or transfer a detector-level guarantee to the VLM.

At threshold 0.9, hold and expected entropy reduction each accept 62 questions, with error fractions of 0.5645 and 0.5323, respectively. APS accepts 65 with error fraction 0.4769. At threshold 0.95, expected entropy reduction retains only eight questions and makes no recorded error, corresponding to 5% coverage. This small accepted subset coincides with the already-correct cases receiving reobservation. Its zero observed error therefore supports a narrow descriptive statement, not a guarantee for the remaining tasks.

For aerospace embodied perception, the practical finding is that changing a sensor state and receiving feedback are necessary elements of an inspection

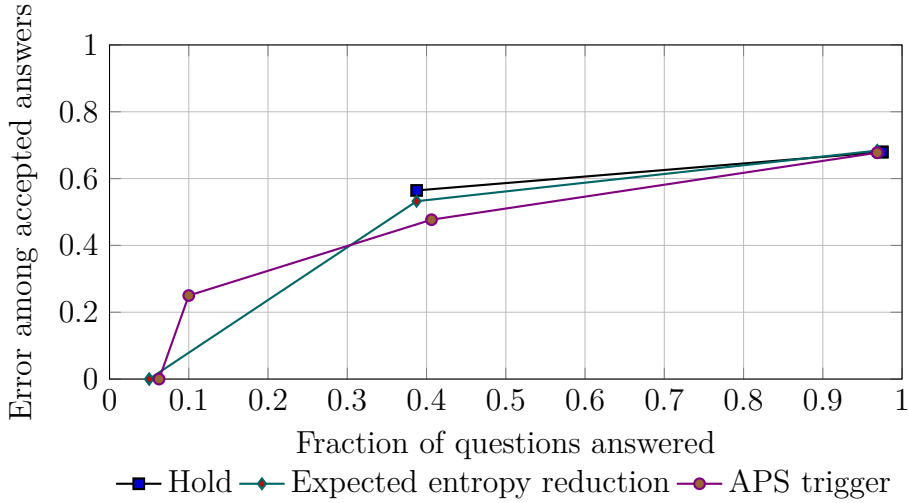


Figure 3: Retrospective error-coverage points at confidence thresholds 0, 0.8, 0.9, 0.95, 0.99, and 1.0, retaining ties. Coverage uses all 160 questions; risk uses the accepted non-abstained answers. Thresholds are diagnostic, with no fitted deployment rule or guarantee.

loop but do not establish decision value. In this setting, the camera ladder changes evidence, offline allocation extracts some useful corrections, and the online trigger often spends its budget on questions that were already resolved. These three observations identify where the evidence chain succeeds and where it stops. They motivate task-conditioned sensing evaluation with verified costs, without supporting broader claims about flight safety, aircraft control, swarms, or energy-aware autonomy.

The principal limitations concern both sensing and execution. The perception backbone has prior event exposure, the source imagery is two-dimensional, the two image dates undergo different degradation, and unmatched buildings receive a default class. Execution provenance is incomplete, only one online seed is recorded, and action budgets differ. Only three evaluation events are represented. A convincing extension requires a recoverable frozen implementation, strict checkpoint and event accounting, regenerated matched controls, and repeated runs with question-level evidence annotations. Those experiments could strengthen or overturn the current interpretation; their outcomes are not assumed here.

7. Conclusion

This study evaluates budgeted reobservation at two levels: paired building predictions and online disaster-inspection answers. In the recorded imagery

setting, a smaller camera footprint changes the perception endpoint and enables useful selection under a fixed building budget. Calibrated entropy improves macro-F1 over the recorded random allocation, but does not outperform raw entropy, and its improvement does not extend uniformly to probability-quality metrics. The online experiment shows a different result: additional observations produce little change in question correctness and are often concentrated on already-correct damage questions.

The supported conclusion is that observation change, uncertainty change, and task value must be measured separately. A functioning feedback loop and higher answer confidence are insufficient evidence of successful sensing allocation. The present results remain conditional on the exposed backbone and archived execution protocol. Future evaluation should first restore full provenance and budget matching, then test whether task-conditioned evidence requirements improve the allocation of new observations on genuinely unseen events. No such improvement is claimed by the current experiments.

Declarations for author completion

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Declaration of competing interest. *To be confirmed by all authors.*

Data and code availability. The working package contains manuscript source and scripts that reconstruct the reported tables from archived predictions and episode logs. Access to the underlying imagery remains subject to its original terms. A public archive, license, anonymization review, and complete data/checkpoint package have not yet been confirmed; a final availability statement must identify the actual release and its restrictions.

Generative AI assistance. Codex was used to assist manuscript restructuring, English drafting, reference checking, and preparation of analysis and typesetting scripts. The present file is an author-review draft. The responsible authors must verify the final content and supply a disclosure consistent with the journal’s current policy; this text does not assert that their verification has already occurred.

Appendix A. Reproducibility and provenance

Appendix A.1. Inputs and reproduction boundary

The manuscript uses a single principal offline population and two online shards. The offline source directory is `runs/benchmarks/paper_cja_mech_v1/`. Paired evaluation predictions are read explicitly from `final_fov/fov_ladder_eval_items.jsonl`; development pairs come from `fov_ladder_val_items.jsonl`; the reported allocation curves are in `budget_allocation.json`. This explicit selection prevents a missing evaluation filename from silently causing a fallback to development results.

The online records are `episodes.jsonl` in two shard directories beneath `runs/benchmarks/cja_agent_vqa/`. Their shared directory-name prefix is `paper_cja_mech_final_shard`, followed by `0of2` or `1of2`. Each configuration/question key must occur once, and each configuration must contain the same 160 question IDs. Recorded correctness is recomputed as a count of the archived flags; this operation does not reannotate images or regenerate model answers.

The local scripts `cja_en/scripts/audit_downloaded_results.py` and `cja_en/scripts/build_assets.py` read these inputs without modifying them or invoking a model. They regenerate the audit, additional analyses, six tables, and two quantitative plots. The audit records input byte counts and full SHA-256 values. The asset generator rejects changed audited inputs and verifies the allocation reconstruction used by the bootstrap. The overview diagram is a schematic of the implemented evaluation, not a synthetic experimental observation.

The paired-building recomputation matches all seven allocation curves at five budgets for accuracy, macro-F1, ECE, Brier score, NLL, executed selections, and net corrections to an absolute tolerance of 10^{-9} . Final three-view metrics also reproduce to that tolerance. These checks establish numerical consistency of available records. They do not certify that the runtime model, original images, or evaluation history can yet be independently recovered.

Appendix A.2. Remaining execution requirements

The frozen manifest was generated on 26 August 2026. Both online shard manifests identify source commit `cfb5c827` (abbreviated) and mark the working tree as dirty. The audit retains its full hash. The currently inspected source is a later revision. Exact reproduction therefore requires the uncommitted runtime changes or an archived source tree, in addition to the named commit. Empty language- and vision-language-model fields must be resolved from contemporaneous logs or author records, rather than filled from current defaults.

The frozen question set and dataset entities are needed to verify their manifest hashes, target construction, scoring rules, and relationships between selection and final populations. Checkpoint files and training manifests are needed to characterize perception exposure directly. Although an additional archive identifies a different event-disjoint perception backend, those results are not combined with the principal tables here: changing the backbone changes the experiment, and its checkpoint provenance remains to be verified.

The online run history also requires repair or regeneration of reference-dependent controls. A safe rerun should bind each dependent episode to the reference question ID, reference action budget, reference result hash, source snapshot, model configuration, and prompt hash. Resuming a completed episode should require those identifiers to match. All configurations should retain requested, allocated, and executed budgets, rather than relying on a single final action total. This is a proposed evaluation correction, not a claim that the existing records satisfy it.

Appendix A.3. Sensitivity to unmatched detections

Table A.1 holds the population fixed to buildings with a matched detection at every view. Its 2,757 buildings exclude all cases assigned a default vector in any of the three observations. The view response remains positive for accuracy and macro-F1, while Brier score and NLL decrease. This analysis checks one potential source of the overall response; it is not a representative subset selected independently of the evaluated detector. It also does not repair possible centroid-matching errors or perception training exposure.

Table A.1: Sensitivity analysis on 2,757 buildings matched in all three recorded views. This selected subset does not replace the full 3,753-building endpoint.

View	Accuracy	Macro-F1	Brier	NLL
Cruise	0.7918	0.7069	0.3367	0.6455
Mid	0.8052	0.7321	0.3176	0.6123
Floor	0.8099	0.7489	0.3175	0.6103

Appendix A.4. Exploratory uncertainty estimates

Table A.2 reports the added paired ROI bootstrap. It conditions on fixed fitted scores, allocations, and three observed events. ROIs are the sampling units within events; individual buildings are not treated as independent draws. With only three events and prior backbone exposure, these intervals cannot establish robustness to unseen disasters. No multiple-comparison-adjusted confirmatory claim is made.

Table A.2: Exploratory paired ROI-bootstrap macro-F1 differences at $b = 0.25$, with 2,000 resamples and seed 20260905.

Paired comparison	Macro-F1 difference	95% interval
Calibrated entropy – Random	+0.0192	[+0.0097, +0.0419]
Calibrated entropy – Raw entropy	-0.0015	[-0.0093, +0.0061]

Appendix A.5. Required extensions, without assumed outcomes

TODO(EXPERIMENT): Repeat the online reference and dependent controls from a frozen execution snapshot with verified per-question budgets and multiple generation seeds. Include a repeated-inference, unchanged-view control to distinguish movement-conditioned evidence from answer variability.

TODO(EXPERIMENT): Evaluate a backbone whose training and tuning events exclude the final events, with recoverable checkpoints, image/ROI manifests, and an explicit rule for unmatched detections. Report matched-only classification alongside an endpoint that penalizes misses, rather than silently changing the original rule.

TODO(EXPERIMENT): Test task-conditioned evidence selection across damage, count, presence, and spatial questions. Record whether the initial answer was wrong, what evidence was acquired, and whether that evidence addressed the unresolved requirement. Compare at matched executed cost, including movement time and inference cost if measured.

TODO(EXPERIMENT): If operational selective answering is claimed, choose thresholds on separate development data and evaluate risk and coverage on held-out tasks. Retrospective confidence plots in this manuscript do not satisfy that design.

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